5.11.3 Constructing a trapezium when its two sides and two angles are given

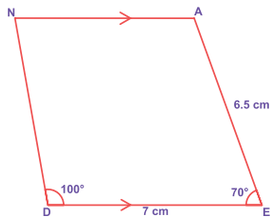
Example 5.28

Construct a trapezium DEAN in which $\overline{DE}$ is parallel to $\overline{NA}$, DE=7*cm*, EA=6.5*cm* $\angle EDN = 100°$ and $\angle DEA = 70°$. Also find its area.

Solution:

Given:

DE=7*cm*, EA=6.5*cm* $\angle EDN = 100°$ and $\angle DEA = 70°$ and $\overline{DE} \parallel \overline{NA}$



Rough diagram

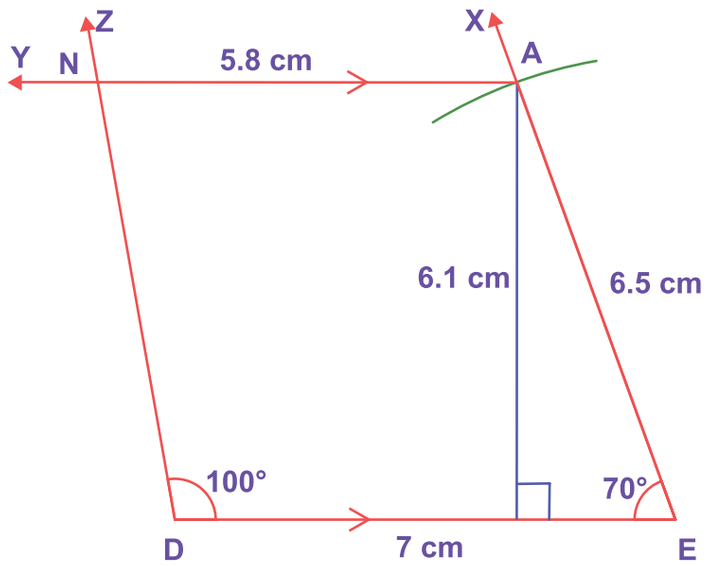


Fig. 5.47

Steps:

1. Draw a line segment DE= 7*cm*.
2. Construct an angle $\angle DEX = 70°$ at E.
3. With E as centre draw an arc of radius 6.5*cm* cutting EX at A.
4. Draw AY parallel to DE.
5. Construct an angle $\angle EDZ = 100°$ at D cutting AY at N.
6. DEAN is the required trapezium.

Calculation of Area:

Area of the trapezium DEAN $= \frac{1}{2} \times h \times (a + b)\, sq.units$

$= \frac{1}{2} \times 6.1 \times (7 + 5.8) = 39.04\, sq.cm$

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